Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3301-WE01

Title: (BEng/MEng)

Structures and Geomatics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

(a) Figure Q1a shows a 4m by 4m in-plan platform that is required to support a uniformly distributed design load (UDL) of 5 kN/m^2 which acts vertically downwards. The platform spans one way between simple supports, which can be assumed to be located at the slab edges (i.e. the span is 4 m exactly).

A possible design for the platform is to assume a set of beams placed side by side, as shown in Figure Q1b. Each beam has a breadth b=250 mm. Considering a **single** beam from the proposed structure in Figure Q2b, determine:

i. the influence line for the bending moment at a point 1 m from one of the supports.
ii. the bending moment at this point due to the UDL applied to the platform,
iii. the bending moment at this point if the UDL of 5 kN/m^2 is confined to the central 2 m of the platform span only.

Hint: You can assume that the loading includes the self-weight of the platform itself.

[40%]

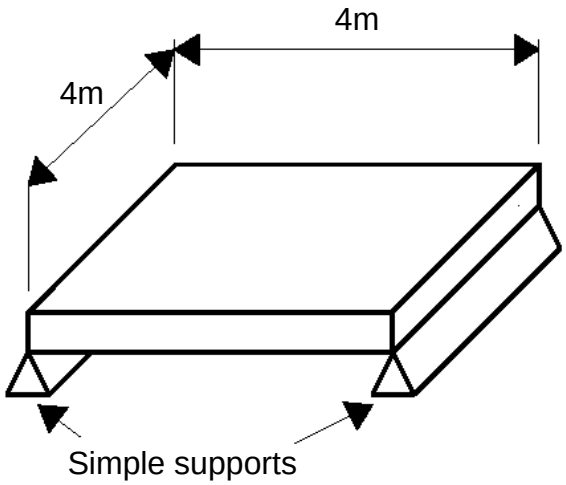


Figure Q1a

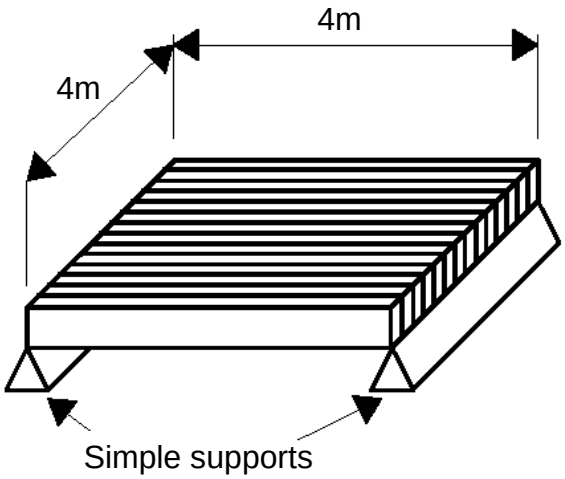


Figure Q1b

continued

(b) Figure Q1c shows a two-member frame. All joints are fixed and connected. Each member has the following properties: Young modulus (E) = 200 GPa, Second moment of area (I) = 420×10^6 mm^4, and Cross-sectional area (A) = 24×10^3 mm^2.

The global degrees of freedom are numbered from 1 to 9, the numbers inside the circles are the joint numbers, and the element (member) numbers are shown inside the squares.

Using the stiffness method, determine the displacements of joint 2 (i.e. D_1, D_2 and D_3, where D_i is the global degree of freedom number i).

Hint: In this question, we are not interested in finding the reactional forces at the supports. Therefore, you can use a simplified technique by constructing only the part of the stiffness matrix that is relevant to the free degrees of freedom.

[60%]

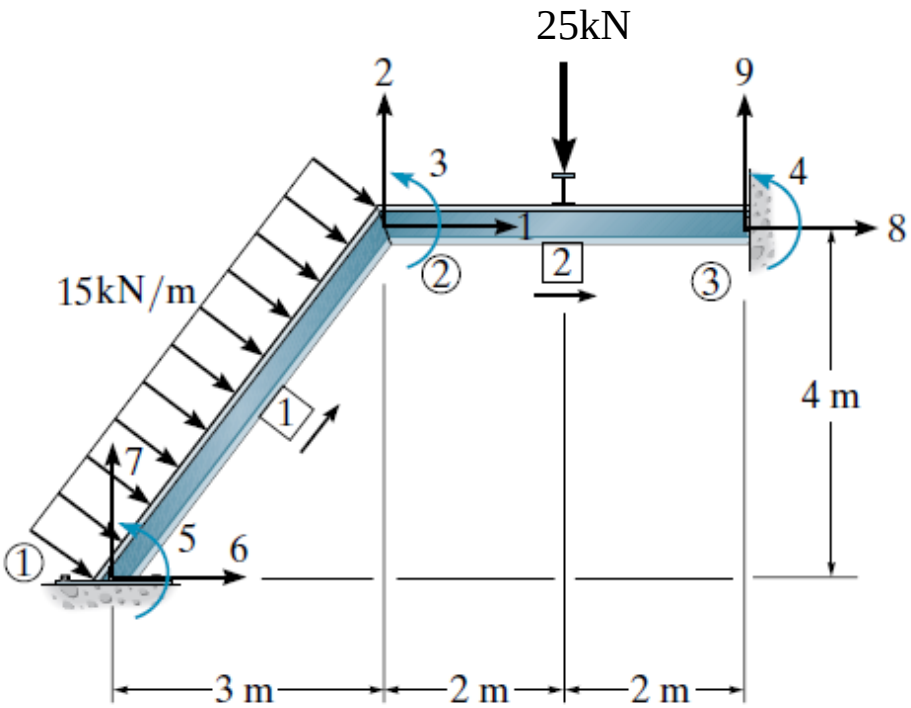


Figure Q1c

Question 2

(a) Figure Q2a shows a continuous beam with two spans (6 m and 4 m) loaded with two concentrated loads P, and 2P, at the centre of each span. Given the plastic moment capacity of the beam, M_p :

i. determine the ultimate value of the load, *P* producing the partial or total collapse of the beam.

ii. determine the bending moment diagram corresponding to the ultimate, collapse load, P.

[50%]

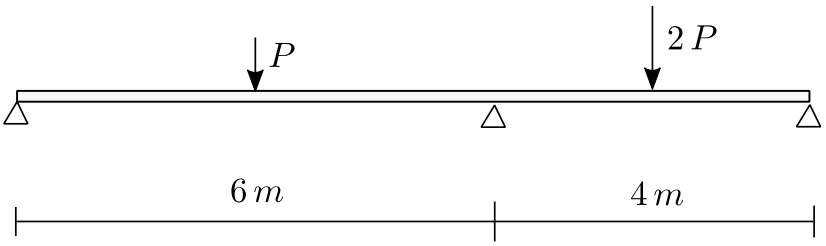


Figure Q2a

(b) Figure Q2b shows a 3m by 5m reinforced concrete slab resting on four columns such that the plate edges are free to rotate but restricted from downward vertical deformations at the supports. The plate is reinforced equally in x- and y-directions with a plastic moment capacity of the slab per unit length, given as M_p.

Determine the ultimate value of the uniformly distributed pressure, *q*, corresponding to the plastic collapse of the slab.

Hint: Consider the collapse mechanism producing the lowest ultimate capacity.

[50%]

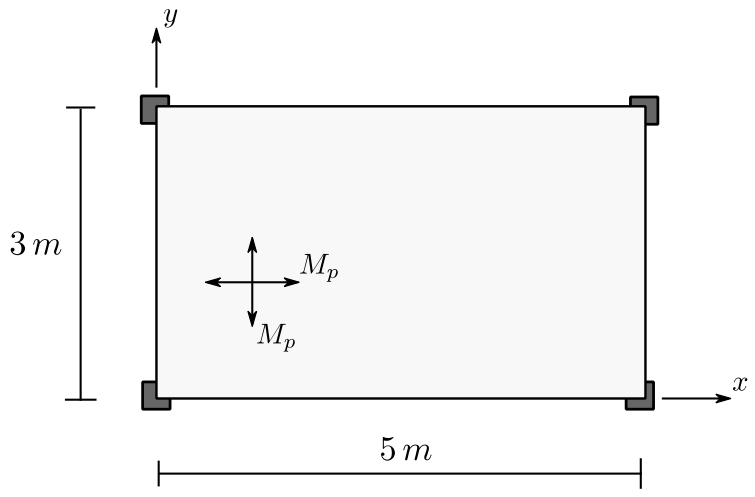


Figure Q2b

continued

Question 3

a) Levelling

Your Site Engineer has observed a line levelling survey between two Ordnance Survey benchmarks and has left you a set of levelling observations, as shown in Table Q3.1. Using the data provided transpose the readings into the correct levelling booking sheet; you may use either method. You have also been asked to complete all the necessary calculations, arithmetic checks and calculate misclosure to ascertain whether the levelling is acceptable. Explain whether the misclosure is acceptable or not and detail some possible sources of error causing unacceptable misclosure and the future mitigation measures required.

Table Q3.1 – Levelling Observations

Instrument Position	Staff Position	Reading
Pavement 1	OS BM on Bridge	0.521m
Pavement 1	End of Bridge Parapet	1.882m
Pavement 2	End of Bridge Parapet	2.647m
Pavement 2	Stone Wall 1	2.548m
Pavement 3	Stone Wall 2	1.625m
Pavement 3	Stone Wall 3	2.673m
Pavement 3	Stone Wall 4	1.277m
Pavement 3	Hedge Line 1	0.518m
Pavement 4	Hedge Line 2	1.432m
Pavement 4	Hedge Line 3	1.785m
Pavement 4	OS BM on Church	1.499m

[40%]

b) Earthwork Volumes

A cutting is to be excavated for an access road to your construction project running through a hill composed of clay soil. The values of ten consecutive cross-sectional areas taken at a constant horizontal interval of 25m are shown in Table Q3.2.

Table Q3.2. Section area data.

	CS1	CS2	CS3	CS4	CS5	CS6	CS7	CS8	CS9	CS10
Chainage(m)	00	25	50	75	100	125	150	175	200	225
Cross sectional area (m^2)	22.4	13.5	30.8	15.8	16.7	14.3	23.2	15.4	16.7	20.3

Once excavated the material will be transported off site in standard ($14m^3$ capacity) wagons. Determine the volume of material that will need to be transported and hence the number of wagon journeys required. Finally, estimate the cost to the project if it costs £106.95/m^3 to excavate the material and £120/wagon to load and remove the material.

[30%]

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c) Geographical Information Systems

You are a Graduate Civil Engineer working for a large consultancy in their design office. Your team is working on the construction of a new housing development as part of this you need to understand what facilities exist in the area, to propose an appropriate number of facilities for the development and to feed this back during a public consultation event. The facilities include: schools, doctors surgeries, playgrounds, local shops, and cafes/restaurants. Explain how you could utilise a Geographical Information System/Model to explore the different data sources and their implications for your development. You should consider what data would need to be included and how these may form layers within your model. You will also need to detail the software functions required to create an output for the public consultation event and discuss what form this output may take.

[30%]

END